

Trade-offs?: Exploring long-run effects of trade liberalisation reforms on local employment, marriage and fertility in Urban India:

A shift-share approach

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Overview:

1. Refresher: research question, data and empirical strategy
2. Relevant context and theory
3. Updated specifications
4. Empirical considerations and robustness checks
5. Results
6. Discussion

Refresher: research question

- **RQ:** What were the long-run effects of 1991 trade liberalisation reforms on local outcomes in Urban India?
- **Outcomes of interest:** employment rates, marriage rates for women at ages 15 - 19, share of women aged 20 - 24 who gave birth in the past year
- Shift-share approach: consider differential impacts of a common shock, dependent on exposure based on pre-determined units
- I follow a **reduced form** approach, and consider changes in outcomes for regions based on **relative degrees** of exposure to tariff reductions

Relevant contextual information

- India's industry-general trade liberalisation reforms in mid-1991 were largely driven by conditional support from the IMF in tackling the balance-of-payments crisis
- Rapid and external implementation lasts until 1997 and move uniformly across industries, after which evidence tariffs are correlated to industry characteristics (Topalova 2010)
- Draws on reduced form analysis and contextual information from Petia Topalova's work, considering poverty and firm productivity from 1991 - 1997

Theory

- Without a general equilibrium model, can't arrive at an aggregate verdict of whether trade liberalisation was beneficial for India
- Effects of trade liberalisation policies on employment (and by gender) in are ambiguously predicted in theories. In the context of a developing country, ex ante hypotheses:
 - ▶ Topalova's findings: regions with lower tariff exposure experienced increased firm productivity - could plausibly expect employment to be lower in regions more protected from trade
 - ▶ Women relatively benefit from trade liberalisation in developing countries: Becker's model of economic discrimination (Gaddis and Pieters 2012), but social structures in India arguably deep-rooted
 - ▶ Converse relationship with changes in men's employment, as households look to smooth labour supply

Data and sample

state_cs_	district_	district_name	year	education_category	rural	
1	25	40	bandarwahi,fazalabad,azamgarh	1991	illiterate	Rural
2	25	40	bandarwahi,fazalabad,azamgarh	2001	illiterate	Rural
3	25	40	bandarwahi,fazalabad,azamgarh	1991	illiterate	Total
4	25	40	bandarwahi,fazalabad,azamgarh	2001	illiterate	Total
5	25	40	bandarwahi,fazalabad,azamgarh	1991	illiterate	Urban
6	25	40	bandarwahi,fazalabad,azamgarh	2001	illiterate	Urban
7	25	40	bandarwahi,fazalabad,azamgarh	1991	Literate but below matrix/second.	Rural
8	25	40	bandarwahi,fazalabad,azamgarh	2001	Literate but below matrix/second.	Rural
9	25	40	bandarwahi,fazalabad,azamgarh	1991	Literate but below matrix/second.	Total
10	25	40	bandarwahi,fazalabad,azamgarh	2001	Literate but below matrix/second.	Total
11	25	40	bandarwahi,fazalabad,azamgarh	1991	Literate but below matrix/second.	Urban
12	25	40	bandarwahi,fazalabad,azamgarh	2001	Literate but below matrix/second.	Urban
13	25	40	bandarwahi,fazalabad,azamgarh	1991	Matrix/secondary but below gra.	Rural
14	25	40	bandarwahi,fazalabad,azamgarh	2001	Matrix/secondary but below gra.	Rural
15	25	40	bandarwahi,fazalabad,azamgarh	1991	Matrix/secondary but below gra.	Total
16	25	40	bandarwahi,fazalabad,azamgarh	2001	Matrix/secondary but below gra.	Total
17	25	40	bandarwahi,fazalabad,azamgarh	1991	Matrix/secondary but below gra.	Urban
18	25	40	bandarwahi,fazalabad,azamgarh	2001	Matrix/secondary but below gra.	Urban
19	25	40	bandarwahi,fazalabad,azamgarh	1991	Graduate and above	Rural
20	25	40	bandarwahi,fazalabad,azamgarh	2001	Graduate and above	Rural
21	25	40	bandarwahi,fazalabad,azamgarh	1991	Graduate and above	Total
22	25	40	bandarwahi,fazalabad,azamgarh	2001	Graduate and above	Total
23	25	40	bandarwahi,fazalabad,azamgarh	1991	Graduate and above	Urban
24	25	40	bandarwahi,fazalabad,azamgarh	2001	Graduate and above	Urban
25	25	40	bandarwahi,fazalabad,azamgarh	1991	Total	Rural
26	25	40	bandarwahi,fazalabad,azamgarh	2001	Total	Rural
27	25	40	bandarwahi,fazalabad,azamgarh	1991	Total	Total
28	25	40	bandarwahi,fazalabad,azamgarh	2001	Total	Total
29	25	40	bandarwahi,fazalabad,azamgarh	1991	Total	Urban
30	25	40	bandarwahi,fazalabad,azamgarh	2001	Total	Urban
31	11	04	bellary,shreega,chitradurga	1991	illiterate	Rural
32	11	04	bellary,shreega,chitradurga	2001	illiterate	Rural

- Analysis is done at the level of composite regions (≈ 380); size closest to district-level with unchanged boundaries from 1991 - 2001
- Restricted to **Urban** observations of regions. Instrument is weak for rural observations
- India cannalized much of commonly grown agricultural produce until 1999, so contrary to many other trade settings, rural production wasn't the most traded.^a

^aMean joint share of workers in agriculture and as cultivators in rural regions: 0.778 (s.d. 0.145)

Refresher: empirical strategy

- OLS: let L denote number of main workers in region d in industry j at time period t

$$Y_{dt} = \alpha + \beta \left(\sum_j \frac{L_{djt}}{L_{dt}} \cdot \text{Tariff}_{jt} \right) + \gamma_d + \lambda_t + \epsilon_{dt}$$

- Issue with OLS: expect industry shares would react to changes in tariffs, likely in a way correlated with the errors.¹
Considering outcomes from 1991 to 2001; 'exogenous' tariff assignment ends after 1997
- **Shift-share instrument:** use lagged pre-treatment employment shares, and instrument for 2001 tariff values with 1997 values. I assume **exogeneity of the shocks**, following Borusyak, Hull, and Jaravel (2018).

¹e.g. Regions with pro-worker labour regulation with more flexible labour market laws might have lower employment rates (Besley and Burgess 2004) and allow for greater movement of workers between industries

Empirical specification (i)

- Reduced form SSIV: Let κ denote a dummy for 2001 observations, 'OS' 'Other Services' industry

$$Y_{dt} = \alpha + \beta \left(\sum_j \frac{L_{dj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \gamma_d + \lambda_t + \kappa \left(\sum_{j \neq OS} \frac{L_{dj1991}}{L_{d1991}} \right) \delta + \epsilon_{dt}$$

- Key assumptions:

- ▶ 1) Relevance: sufficiently strong persistence of industry employment shares, and correlation between 1997 and 2001 tariff values ✓
- ▶ 2) Tariffs are 'as-good-as-randomly' assigned to industries.² Share-weighted average of random shocks is itself random ✓
- ▶ 3) Shocks are large in number: are ten industry groups sufficient? ?

²Actually require a less strong assumption; tariffs are not systematically higher (lower) in industries concentrated in regions which had shocks influencing labour supply, marriage, fertility

Incomplete shares control

- **Issue:** 'Other services' industry group (includes financial intermediation, business activities, education, health services) is assigned a tariff of 0 - this endogenously limits the size of the instrument
- Regional FEs control for effect of the sum of baseline shares varying across regions in the first period
- But also need to control for the differential impact the sum of baseline shares has in the second period³
- **Specification:** interact the sum of baseline shares with a dummy for the second period - equivalent to controlling for a linear time-trend in sum of baseline shares

³Plausible to expect regions with high 'other services' industry (lower instrument value) to have had a higher change in error term over time, through greater productivity and human capital, hence there would be bias

Empirical specification (ii)

- Considering gender-specific components of shocks
- Multiply start-of-period industry employment share with the start-of-period gender share for that industry - similar to Autor et al (2019)
- Let f denote number of women, m denote number of men:

$$Y_{dt} = \alpha + \beta_1 \left(\sum_j \frac{L_{dfj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \beta_2 \left(\sum_j \frac{L_{dmj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) \\ + \gamma_d + \lambda_t + \kappa \left(\sum_{j \neq OS} \frac{L_{dfj1991}}{L_{d1991}} \right) \delta_1 + \kappa \left(\sum_{j \neq OS} \frac{L_{dmj1991}}{L_{d1991}} \right) \delta_2 + \epsilon_{dt}$$

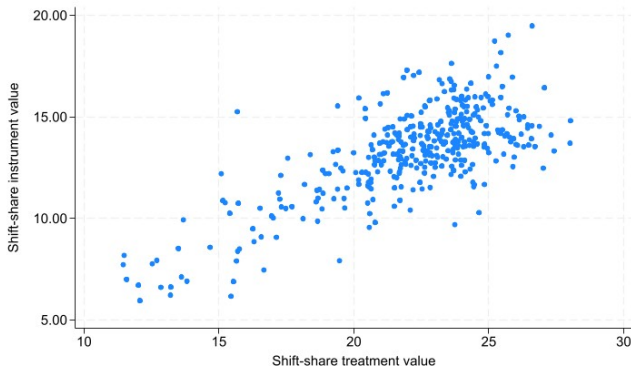
Exposure-robust standard errors

- **Issue:** Regions with similar industry employment composition mechanically have similar instrument values, and so likely correlated error terms
- Borusyak, Hull and Jaravel show estimating the regression at the shock-level (industry) yields identical coefficients but corrected SEs - designed a command that helps implement
- `ssaggregate` transforms the dataset from location-level to industry-level, constructing share-weighted averages of variables⁴
- Let $\bar{y}$ be partialled-out, share-weighted variable. Reduced form where $Tariff_{1991,1997}$ instruments for transformed treatment $\bar{x}_{jt}$

$$\bar{Y}_{jt} = \alpha + \beta \text{Tariff}_{j1991,1997} + \lambda_t + \epsilon_{jt} \quad (1)$$

⁴The regression run is now weighted by the shares across locations - i.e. let industry-employment share for industry j in location d be w_{dj} . Each variable in the industry-level regression is weighted by $W_d = \frac{1}{D} \sum_d w_{dj}$ - the average exposure of locations to industry j

Relevant first stage for urban sample



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⁵Scatter plot showing values for 2001 observations only (instrument and treatment values identical for 1991). Urban sample: First-stage F-stat of 289.97. Gendered split: female-specific instrument = 36.36, male-specific = 211.29. Mean of treatment = 30.3 (9.66), mean of instrument = 25.9 (13.5)

Results: employment

	(1) Emp (Overall)	(2) Female emp	(3) Male emp	(4) Emp (Overall log)	(5) Female emp (Log)	(6) Male emp (Log)
SSIV	0.007 (0.009)	0.010 (0.010)	0.006 (0.012)	0.020 (0.032)	0.177** (0.085)	0.011 (0.026)
Log pop				1.01 (0.053)		
Log female pop					1.04 (0.189)	
Log male pop						1.012 (0.014)
Level of estimation	Location	Location	Location	Location	Location	Location
N	382	382	382	382	382	382

Clustered standard errors, at composite region level, in parantheses

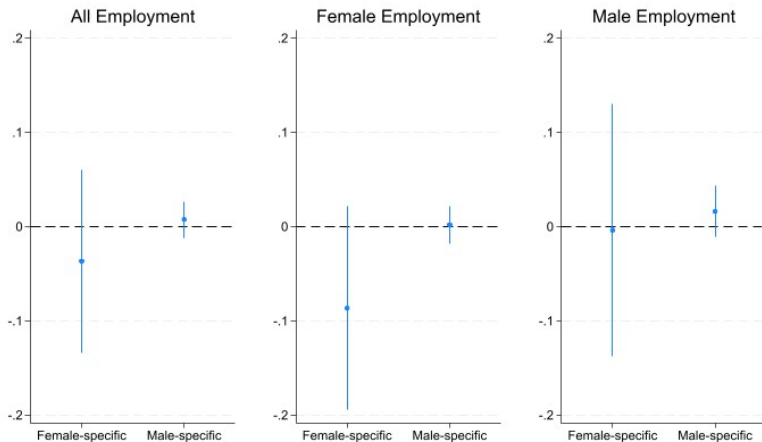
*** p<0.01, ** p<0.05, * <0.1

SSIV results for effect on employment (resized coefficients by s.f10)

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⁶For ease of interpretability, resized coefficients by multiple of 10 (i.e. run regression on $\frac{x}{10}$). Mean of employment: 0.284 (0.042), female employment: 0.089 (0.048), male employment: 0.459 (0.053). Industry-level regressions provide somewhat different coefficients, but also very small in magnitude and no statistical significance.

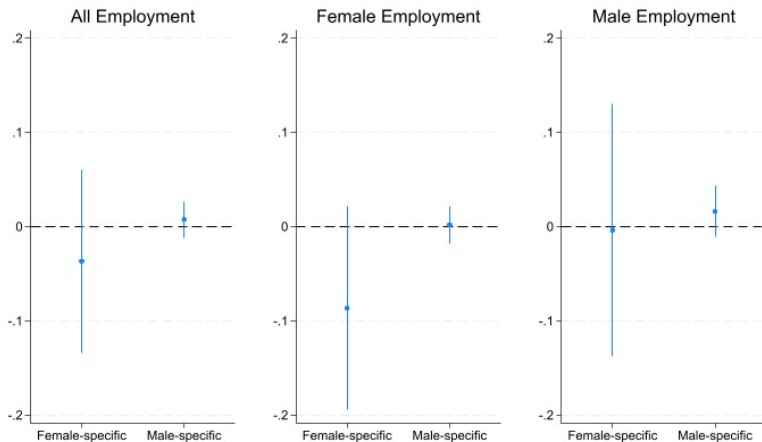
Results: employment, by gender-specific shock



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⁷Resized coefficient by multiple of 10. From decomposed gender-specific SSIV

Results: log employment, by gender-specific shock



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⁸Log(number of employed) on LHS, and log(population) on RHS

Results: fertility and marriage

	(1) Marriage 15 - 19	(2) Share of births 20 - 24	(3) Marriage 15 - 19	(4) Share of births 20 - 24
SSIV	0.003 (0.021)	0.000 (0.019)	-0.002 (0.017)	0.001 (0.003)
Level of estimation	Location	Location	Industry	Industry
N	376	376	9	9

Clustered standard errors at regional level (1, 2). Exposure-robust standard errors (3, 4).

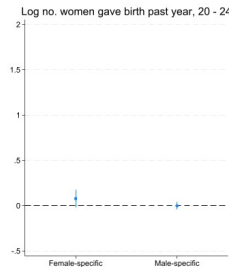
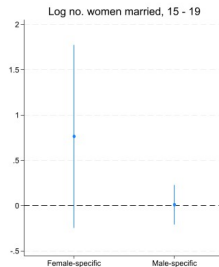
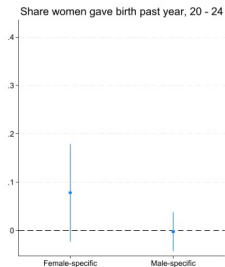
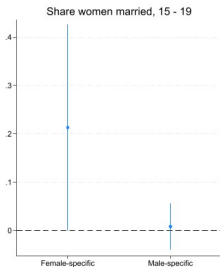
*** p<0.01, ** p<0.05, * p<0.1

SSIV results for effect on marriage and fertility rates (resized coefficients by
s.f.10)

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⁹Notes: Mean proportion of 15 - 19 year old women married: 0.198 (0.100), mean share of 20 - 24 year old women who gave birth in past year: 0.121 (0.058). Log(level) results do not substantially differ by coefficient or significance.

Results: fertility and marriage, by gender specific shock



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¹⁰Resized coefficients by multiple of 10 - regression run on $\frac{x}{10}$

What do the null results tell us?

- By gender-specific shocks, find negative but statistically insignificant coefficient on female-specific share for female employment; more protected a region is to trade, the lower female employment
- A 10-unit (approx 1 s.d) increase in a region's degree of protection to trade corresponds to a 21.3% point increase in share of 15 - 19 year old women married.
- **Why don't we see significant effects on employment?**
Employment within a region across time doesn't actually vary substantially - this most likely captures a more formal measure of employment.
- Economic growth in India post-reform did not absorb growing workforce and facilitate structural transformation (Bhagwati and Panagariya 2013)

Jackknife resampling (/leave-one-out construction of shifts)

- Limitation in my approach: 10 industry groups in the census. Shift-share assumption: shocks are large in number, so this is not ideal
- Jackknife resampling: iteratively leaving out one industry group and then estimating on the remaining sample
- Are my results sensitive?

Things to work on

- Consistency of results
- Improving measure of employment:
 - ▶ Have data on main activities of non-workers; includes household activities, students, retirees, inmates (discount under 15, students and retirees from employment denominator)
- Explaining lack of significant results: How does what I find tie in with the literature?
 - ▶ Can I test or find support for my explanations? Framing empirical findings through a model

References

-  Besley, Timothy and Robin Burgess (2004). “Can labor regulation hinder economic performance? Evidence from India”. In: *The Quarterly Journal of Economics*.
-  Bhagwati, Jagdish and Arvind Panagariya (2013). *Reforms and Economic Transformation in India*. Oxford University Press.
-  Borusyak, Kirill, Peter Hull, and Xavier Jaravel (2018). “Quasi-experimental shift-share research designs”. In: *NBER*.
-  Gaddis, Isis and Janneke Pieters (2012). “Trade Liberalization and Female Labor Force Participation: Evidence from Brazil”. In: *Institute for the Study of Labor IZA*.
-  Topalova, Petia (2010). “Factor Immobility and Regional Impacts of Trade Liberalization: Evidence on Poverty from India”. In: *American Economic Journal: Applied Economics*.